

Up next in Hands-on Labs →

Lab 03: Artifacts

 Hands-on lab · 45 min · Env: GitHub + Docker · Overview & prereqs · Download PDF ·

Reset checks

CI/CD ENGINEERING · KNOWLEDGE BASE

Lab 02: Build & Test

CICD BY Nabawy

Reader editionNabawy

Objective

Split Lab 01's single job into a fast, cached PR pipeline: hermetic build job → unit test job, total under 10 minutes.

Prerequisites

- Lab 01 green; concepts: Build Systems, Testing Strategy.

Architecture

PR → job build (npm ci, cache on lockfile) → job test (needs: build, unit suite) → combined status check.

Setup

Add a second test file so the suite has something real to run.

Step 1 — Split and cache

Execute

```
jobs: build: runs-on: ubuntu-latest steps: - uses: actions/checkout@v4 - uses: actions/setup-node@v4 with: { node-version: 20, cache: npm } - run: npm ci test: needs: build runs-on: ubuntu-latest steps: - uses: actions/checkout@v4 - uses: actions/setup-node@v4 with: { node-version: 20, cache: npm } - run: npm ci - run: npm test
```

Expected Result

Two jobs, test waits for build; reruns are faster (warm npm cache).

Why

needs: is the DAG edge; cache keyed on the lockfile is the hermetic speedup (see pipelines DAG).

Step 2 — Prove hermeticity

Execute

Delete `node_modules` locally, run `npm ci && npm test` — same result as CI.

Expected Result

Identical pass/fail locally and on the runner.

Why

Lockfile-exact install means the runner builds what you built — no ambient-machine surprises.

Verification

- [] Two jobs visible, ordered by needs.
- [] Cache hit shown on rerun.
- [] Local `npm ci` reproduces CI result.

Failure Scenarios

- Cache never hits: lockfile not committed or key inputs changed — verify `package-lock.json` is tracked.
- `test` runs before build finishes: missing `needs:` — add the edge.

Cleanup

Keep; Lab 03 packages this build's output.

What This Proved

DAG ordering, hermetic installs, lockfile caching. Continue with Lab 03.

Interview Notes

- What `needs:` actually guarantees — and the failure you get when the edge is missing.
- Why the cache key includes the lockfile, and what "cache never hits" tells you about the repo.

RELATED BOOKS



Lab 03: Artifacts

45 min · INT

→ Next in path



Lab 01: First Pipeline

45 min · BEG

→ Previous in path



Build Systems

30 min · INT

→ Prerequisite



Lab 01: GitHub Webhook Trigger for Jenkins

45 min · INT

→ Same path